**Topic**: Laplace's equation

**Question**: Find a solution to the partial differential equation.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \text{ for } 0 \leq x \leq 1 \text{ and } 0 \leq y \leq 2$$

$$u(x,0) = \sin(\pi x) \qquad u(0,y) = 0$$

$$u(x,2) = 0 \qquad u(1,y) = 0$$

**Answer choices**:

A $u(x,y) = \dfrac{1}{\sinh 2\pi}\sinh(\pi(1-x))\sin(\pi y)$

B $u(x,y) = \displaystyle\sum_{n=1}^{\infty}(1-(-1)^n)\sinh(n\pi(2-y))\sin(n\pi x)$

C $u(x,y) = \dfrac{1}{\sinh 2\pi}\sinh(\pi(2-y))\sin(\pi x)$

D $u(x,y) = \displaystyle\sum_{n=1}^{\infty}(1-(-1)^n)\sinh(n\pi(1-x))\sin(n\pi y)$

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**Solution**: C

If we find the first and second derivatives of the product solution,

$$u(x,y) = v(x)w(y)$$

$$\frac{\partial u}{\partial x} = \frac{dv}{dx}w \qquad \frac{\partial u}{\partial y} = v\frac{dw}{dy}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{d^2v}{dx^2}w \qquad \frac{\partial^2 u}{\partial y^2} = v\frac{d^2w}{dy^2}$$

and then plug the second derivatives into Laplace's equation,

$$\frac{d^2v}{dx^2}w + v\frac{d^2w}{dy^2} = 0$$

then we can separate variables.

$$\frac{d^2v}{dx^2}w = -v\frac{d^2w}{dy^2}$$

$$\left(\frac{1}{v}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w}\right)\frac{d^2w}{dy^2}$$

Set both sides of the equation equal to the separation constant.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

Break this into two separate ordinary differential equations.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\lambda$$

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$$\frac{d^2v}{dx^2} = -\lambda v(x)$$

$$\frac{d^2v}{dx^2} + \lambda v = 0$$

and

$$-\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

$$\frac{d^2w}{dy^2} = \lambda w(y)$$

$$\frac{d^2w}{dy^2} - \lambda w = 0$$

Our boundary conditions are $w(2) = 0$, $v(0) = 0$, and $v(1) = 0$, and the solution to the first ordinary differential equation boundary value problem is

$$\lambda_n = \left(\frac{n\pi}{1}\right)^2 = (n\pi)^2 \qquad v_n(x) = C\sin(n\pi x) \qquad n = 1, 2, 3, \ldots$$

When we plug this value for $\lambda$ into the second equation, we get

$$\frac{d^2w}{dy^2} - (n\pi)^2 w = 0$$

$$w(2) = 0$$

Because $\lambda > 0$, we know that the solution to this second order equation will be

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$$w(y) = c_1 e^{-\sqrt{\lambda}y} + c_2 e^{\sqrt{\lambda}y}$$

Equivalently, this can be rewritten as

$$w(y) = c_1 \sinh(\sqrt{\lambda}(y-2)) + c_2 \cosh(\sqrt{\lambda}(y-2))$$

which allows us to apply $w(2) = 0$. When we do, we find $c_2 = 0$, and the solution is

$$w(y) = c_1 \sinh(n\pi(y-2))$$

Then the product solution is

$$u_1(x,y) = v(x)w(y)$$

$$u(x,y) = \sum_{n=1}^{\infty} B_n \sinh(n\pi(y-2))\sin(n\pi x) \qquad n = 1, 2, 3, \ldots$$

Substitute $u(x,0) = \sin(\pi x)$.

$$u(x,0) = \sum_{n=1}^{\infty} B_n \sinh(-2n\pi)\sin(n\pi x) = \sin(\pi x)$$

$$B_1 = -\frac{1}{\sinh 2\pi}$$

$$B_2 = B_3 = \ldots = 0$$

Then the solution is

$$u(x,y) = \frac{1}{\sinh 2\pi} \sinh(\pi(2-y))\sin(\pi x)$$

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**Topic**: Laplace's equation

**Question**: Find a solution to the partial differential equation.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \text{ for } 0 \leq x \leq \pi \text{ and } 0 \leq y \leq \pi$$

$$u(x,0) = 0 \qquad u(0,y) = 1$$

$$u(x,\pi) = 0 \qquad u(\pi,y) = \sin 2y$$

**Answer choices**:

A $$\sum_{n=1}^{\infty} \frac{2}{n\pi \sinh(n\pi)}(1-(-1)^n)\sinh(n(\pi-x))\sin(ny) + \frac{1}{\sinh 2\pi}\sin(2x)\sinh(2y)$$

B $$\sum_{n=1}^{\infty} \frac{2}{n\pi \sinh(n\pi)}(1-(-1)^n)\sinh(n(\pi-x))\sin(ny) + \frac{1}{\sinh 2\pi}\sinh(2x)\sin(2y)$$

C $$\sum_{n=1}^{\infty} \frac{2}{n\pi \sinh(n\pi)}(1-(-1)^n)\sin(nx)\sinh(n(\pi-y)) + \frac{1}{\sinh 2\pi}\sin(2x)\sinh(2y)$$

D $$\sum_{n=1}^{\infty} \frac{2}{n\pi \sinh(n\pi)}(1-(-1)^n)\sin(nx)\sinh(n(\pi-y)) + \frac{1}{\sinh 2\pi}\sinh(2x)\sin(2y)$$

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**Solution**: B

We can say that $u = u_1 + u_2$, where $u_1$ is the solution of

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$u(x,0) = 0 \qquad u(0,y) = 1$$

$$u(x,\pi) = 0 \qquad u(\pi,y) = 0$$

and $u_2$ is the solution of

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$u(x,0) = 0 \qquad u(0,y) = 0$$

$$u(x,\pi) = 0 \qquad u(\pi,y) = \sin 2y$$

Lets start with $u_1$. If we find the first and second derivatives of the product solution,

$$u(x,y) = v(x)w(y)$$

$$\frac{\partial u}{\partial x} = \frac{dv}{dx}w \qquad \frac{\partial u}{\partial y} = v\frac{dw}{dy}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{d^2 v}{dx^2}w \qquad \frac{\partial^2 u}{\partial y^2} = v\frac{d^2 w}{dy^2}$$

and then plug the second derivatives into Laplace's equation,

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$$\frac{d^2v}{dx^2}w + v\frac{d^2w}{dy^2} = 0$$

then we can separate variables.

$$\frac{d^2v}{dx^2}w = -v\frac{d^2w}{dy^2}$$

$$\left(\frac{1}{v}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w}\right)\frac{d^2w}{dy^2}$$

Set both sides of the equation equal to the separation constant.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = \lambda$$

Break this into two separate ordinary differential equations.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = \lambda$$

$$\frac{d^2v}{dx^2} = \lambda v(x)$$

$$\frac{d^2v}{dx^2} - \lambda v = 0$$

and

$$-\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = \lambda$$

$$\frac{d^2w}{dy^2} = -\lambda w(y)$$

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$$\frac{d^2w}{dy^2} + \lambda w = 0$$

Our boundary conditions are $w(\pi) = 0$, $w(0) = 0$, and $v(\pi) = 0$, and the solution to the second ordinary differential equation boundary value problem is

$$\lambda_n = \left(\frac{n\pi}{\pi}\right)^2 = n^2 \qquad w_n(y) = C\sin(ny) \qquad n = 1, 2, 3, \ldots$$

When we plug this value for $\lambda$ into the second equation, we get

$$\frac{d^2v}{dx^2} - n^2 v = 0$$

$$v(\pi) = 0$$

Because $\lambda > 0$, we know that the solution to this second order equation will be

$$v_n(x) = c_1 \sinh nx + c_2 \cosh nx$$

Equivalently, this can be rewritten as

$$v_n(x) = c_1 \sinh n(x - \pi) + c_2 \cosh n(x - \pi)$$

which allows us to apply $v(\pi) = 0$. When we do, we find $c_2 = 0$, and the solution is

$$v_n(x) = c_1 \sinh n(x - \pi)$$

Then the solution for $u_1$ is

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$$u_1(x, y) = v_n(x)w_n(y)$$

$$u_1(x, y) = \sum_{n=1}^{\infty} B_n \sinh(n(x - \pi))\sin(ny) \qquad n = 1, 2, 3, \ldots$$

Substitute $u(0,y) = 1$.

$$u_1(0,y) = \sum_{n=1}^{\infty} B_n \sinh(n(0 - \pi))\sin(ny) = 1$$

$$B_n \sinh(-n\pi)\sin(ny) = \frac{2}{\pi}\int_0^{\pi} \sin(ny)dy = \frac{2}{n\pi}(-\cos(ny))\Big|_0^{\pi} = \frac{2}{n\pi}(1 - (-1)^n)$$

$$u_1(x, y) = \sum_{n=1}^{\infty} \frac{2}{n\pi \sinh(n\pi)}(1 - (-1)^n)\sinh(n(\pi - x))\sin(ny)$$

Solve for $u_2$.

$$\frac{d^2v}{dx^2}w + v\frac{d^2w}{dy^2} = 0$$

$$\frac{d^2v}{dx^2} - \lambda v = 0$$

and

$$\frac{d^2w}{dy^2} + \lambda w = 0$$

Our boundary conditions are $w(\pi) = 0$, $w(0) = 0$, and $v(\pi) = 0$, and the solution to the second ordinary differential equation boundary value problem is

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$$\lambda_n = \left(\frac{n\pi}{\pi}\right)^2 = n^2 \qquad w_n(y) = C\sin(ny) \qquad n = 1, 2, 3, \ldots$$

When we plug this value for $\lambda$ into the second equation, we get

$$\frac{d^2v}{dx^2} - n^2v = 0$$

$$v(\pi) = 0$$

Because $\lambda > 0$, we know that the solution to this second order equation will be

$$v_n(x) = c_1 \sinh nx + c_2 \cosh nx$$

which allows us to apply $v(\pi) = 0$. When we do, we find $c_2 = 0$, and the solution is

$$v_n(x) = c_1 \sinh n(x - \pi)$$

Then solution for $u_2$ is

$$u_2(x, y) = v_n(x)w_n(y)$$

$$u_2(x, y) = \sum_{n=1}^{\infty} B_n \sinh(nx)\sin(ny) \qquad n = 1, 2, 3, \ldots$$

Substitute $u(\pi, y) = \sin 2y$.

$$u_2(\pi, y) = \sum_{n=1}^{\infty} B_n \sinh(n\pi)\sin(ny) = \sin 2y$$

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$$B_2 = \frac{1}{\sinh 2\pi}$$

$$u_2(x,y) = \frac{1}{\sinh 2\pi}\sinh(2x)\sin(2y)$$

Then the solution is

$$u = u_1 + u_2$$

$$u(x,y) = \sum_{n=1}^{\infty} \frac{2}{n\pi \sinh(n\pi)}(1-(-1)^n)\sinh(n(\pi - x))\sin(ny) + \frac{1}{\sinh 2\pi}\sinh(2x)\sin(2y)$$

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**Topic**: Laplace's equation

**Question**: Find a solution to the partial differential equation.

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \text{ for } 0 \leq x \leq 2 \text{ and } 0 \leq y \leq 5$$

$$u(x,0) = 0 \qquad u(0,y) = 0$$

$$u(x,5) = \sin(\pi x) + \sin(2\pi x) \qquad u(2,y) = 0$$

**Answer choices**:

A $$u(x,y) = \frac{1}{\sinh 5\pi}\sinh(\pi y)\sin(\pi x) + \frac{1}{\sinh 10\pi}\sinh(2\pi y)\sin(2\pi x)$$

B $$u(x,y) = \frac{1}{\sinh 5\pi}\sin(\pi y)\sinh(\pi x) + \frac{1}{\sinh 10\pi}\sin(2\pi y)\sinh(2\pi x)$$

C $$u(x,y) = \frac{1}{\sinh 5\pi}\sin(\pi x) + \frac{1}{\sinh 10\pi}\sin(2\pi x)$$

D $$u(x,y) = \frac{1}{\sin 10\pi}\sinh(\pi y)\sin(\pi x) + \frac{1}{\sin 20\pi}\sinh(2\pi y)\sin(2\pi x)$$

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**Solution**: A

If we find the first and second derivatives of the product solution,

$$u(x,y) = v(x)w(y)$$

$$\frac{\partial u}{\partial x} = \frac{dv}{dx}w \qquad \frac{\partial u}{\partial y} = v\frac{dw}{dy}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{d^2v}{dx^2}w \qquad \frac{\partial^2 u}{\partial y^2} = v\frac{d^2w}{dy^2}$$

and then plug the second derivatives into Laplace's equation,

$$\frac{d^2v}{dx^2}w + v\frac{d^2w}{dy^2} = 0$$

then we can separate variables.

$$\frac{d^2v}{dx^2}w = -v\frac{d^2w}{dy^2}$$

$$\left(\frac{1}{v}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w}\right)\frac{d^2w}{dy^2}$$

Set both sides of the equation equal to the separation constant.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

Break this into two separate ordinary differential equations.

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\lambda$$

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$$\frac{d^2v}{dx^2} = -\lambda v(x)$$

$$\frac{d^2v}{dx^2} + \lambda v = 0$$

and

$$-\left(\frac{1}{w(y)}\right)\frac{d^2w}{dy^2} = -\lambda$$

$$\frac{d^2w}{dy^2} = \lambda w(y)$$

$$\frac{d^2w}{dy^2} - \lambda w = 0$$

Our boundary conditions are $w(0) = 0$, $v(0) = 0$, and $v(2) = 0$, and the solution to the first ordinary differential equation boundary value problem is

$$\lambda_n = \left(\frac{n\pi}{2}\right)^2 \qquad v_n(x) = C\sin\left(\frac{n\pi}{2}\right) \qquad n = 1, 2, 3, \ldots$$

When we plug this value for $\lambda$ into the second equation, we get

$$\frac{d^2w}{dy^2} - \left(\frac{n\pi}{2}\right)^2 w = 0$$

$$w(2) = 0$$

Because $\lambda > 0$, we know that the solution to this second order equation will be

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$$w(y) = c_1 \sinh\left(\frac{n\pi y}{2}\right) + c_2 \cosh\left(\frac{n\pi y}{2}\right)$$

which allows us to apply $w(0) = 0$. When we do, we find $c_2 = 0$, and the solution is

$$w(y) = c_1 \sinh\left(\frac{n\pi y}{2}\right)$$

Then the product solution is

$$u_1(x, y) = v(x)w(y)$$

$$u(x, y) = \sum_{n=1}^{\infty} B_n \sinh\left(\frac{n\pi y}{2}\right) \sin\left(\frac{n\pi x}{2}\right) \qquad n = 1, 2, 3, \ldots$$

Substitute $u(x,5) = \sin(\pi x) + \sin(2\pi x)$.

$$u(x,5) = \sum_{n=1}^{\infty} B_n \sinh\left(\frac{5n\pi}{2}\right) \sin\left(\frac{n\pi x}{2}\right) = \sin(\pi x) + \sin(2\pi x)$$

$$B_2 = \frac{1}{\sinh 5\pi}$$

$$B_4 = \frac{1}{\sinh 10\pi}$$

$$B_1 = B_3 = \ldots = 0$$

Then the solution is

$$u(x, y) = \frac{1}{\sinh 5\pi} \sinh(\pi y)\sin(\pi x) + \frac{1}{\sinh 10\pi} \sinh(2\pi y)\sin(2\pi x)$$

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